

Solution Code - Engineering concepts - Algorithms_DataStructures - Library_Management_System

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: LibrarySearch.java

```
1 | import java.util.Arrays;
2 |
3 | class Book implements Comparable<Book> {
4 |     String title;
5 |     public Book(String title) { this.title = title; }
6 |     public int compareTo(Book o) { return this.title.compareTo(o.title); }
7 | }
8 |
9 | public class LibrarySearch {
10 |     public static int linearSearch(Book[] arr, String title) {
11 |         for (int i = 0; i < arr.length; i++) if (arr[i].title.equalsIgnoreCase(title)) return i;
12 |         return -1;
13 |     }
14 |     public static int binarySearch(Book[] arr, String title) {
15 |         int l = 0, r = arr.length - 1;
16 |         while (l <= r) {
17 |             int m = l + (r - l) / 2;
18 |             int cmp = arr[m].title.compareToIgnoreCase(title);
19 |             if (cmp == 0) return m;
20 |             if (cmp < 0) l = m + 1;
21 |             else r = m - 1;
22 |         }
23 |         return -1;
24 |     }
25 | }
```